

```
#include <TinyGPS++.h>
```

```
uint32_t tsLastReport = 0;
```

```
// GPS variables
```

```
String textMessage;
```

```
int Number_of_SATS;
```

```
String latitude="0.0";
```

```
String longitude="0.0";
```

```
TinyGPSPlus gps;
```

```
// Buzzer and button pins
```

```
int buz = D0;
```

```
int btn = D1;
```

```
// Geo-fencing coordinates (static coordinates)
```

```
double RESTRICTED_LAT = 0.0; // Example: Bangalore coordinates
```

```
double RESTRICTED_LON = 0.0; // Change these to your restricted area
```

```
const double GEO_FENCE_RADIUS = 50.0; // 100 meters radius
```

```
// Variables for speed calculation
```

```
bool isBuzzerOn = false;
```

```
bool geoset=false;
```

```
bool geosmsset=true;
```

```
void setup() {
```

```
    Serial.begin(9600);
```

```
    pinMode(buz, OUTPUT);
```

```
    digitalWrite(buz, LOW);
```

```
    pinMode(btn, INPUT_PULLUP);
```

```
    while(millis()<30000){
```

```
        getGPS();
```

```
    }
```

```
    RESTRICTED_LAT=gps.location.lat();
```

```
    RESTRICTED_LON=gps.location.lng();
```

```
}
```

```
void loop() {
```

```
    getGPS();
```

```
    getGPS();
```

```
    // Check emergency button
```

```

if (!digitalRead(btn)) {
    digitalWrite(buz, HIGH);

    getGPS();

    String str = String("Person X is in Emergency Situation his live location=
http://maps.google.com/maps?q=" + latitude + "," + longitude);

    sendSMS(str);

    for (int i = 0; i < 200; i++) {

        delay(100);
    }

    digitalWrite(buz, LOW);

}

// Check geo-fencing
checkGeoFencing();

delay(10);
}

void getGPS() {
    while (Serial.available() > 0) {
        gps.encode(Serial.read());

        latitude = String(gps.location.lat(), 6);

```

```

longitude = String(gps.location.lng(), 6);

}

}

void checkGeoFencing() {
    if (gps.location.isValid()) {
        double currentLat = gps.location.lat();
        double currentLon = gps.location.lng();

        // Calculate distance to restricted area

        double distance = TinyGPSPlus::distanceBetween(currentLat, currentLon,
REstricted_LAT, Restricted_LON);

        if (distance > GEO_FENCE_RADIUS && geosmsset) {
            geosmsset=false;
            if (!isBuzzerOn) {
                digitalWrite(buz, HIGH);
                isBuzzerOn = true;
                geoset=true;
            //    Serial.println("ALERT: Entered restricted area!");

            // Optional: Send SMS for geo-fence violation

            String geoAlert = String("Entered restricted area! Distance: " + String(distance) + "m
at http://maps.google.com/maps?q=" + latitude + "," + longitude);

            sendSMS(geoAlert);

            digitalWrite(buz,LOW);
        }
    }
}

```

```
}  
}  
}
```

```
void sendSMS(String message) {  
  Serial.println("AT+CMGF=1");  
  delay(1000);  
  Serial.println("AT+CMGS=\"+918088735075\"\\r");  
  delay(1000);  
  Serial.println(message);  
  delay(100);  
  Serial.println((char)26);  
  delay(8000);  
}
```